

# STATEMENT OF PURPOSE

**Applicant:** Xuan Thanh Trinh

**Program:** Ph.D. in Electrical, Electronics and Computer Engineering

**Proposed Advisor:** Assistant Professor Van-Hai Bui

**Institution:** University of Michigan-Dearborn

## 1. Introduction

As a final-year Electrical Engineering student at Hanoi University of Science and Technology (HUST), my research at the Power Grid and Renewable Energy (PGRE) Laboratory focuses on developing distributed mathematical optimization models for decentralized energy management. While this foundation drove my graduation thesis on microgrid resilience, I recognized that classical optimization alone cannot scale to the real-time computational demands of modern power systems. To overcome these computational bottlenecks, I require advanced training in scalable, cyber-physical control systems. The Ph.D. program at UM-Dearborn perfectly aligns with this need, providing the rigorous framework I need to become a highly specialized research engineer capable of designing decentralized architectures for high renewable penetrations.

## 2. Academic Background

Ranking top 3 in HUST's Advanced Program in Electrical Engineering and Renewable Energy instilled in me a systems-level perspective. Rather than isolating individual grid components, I trained to deconstruct complex power networks into dynamic, tractable optimization problems. This mathematical foundation drives my independent research trajectory, allowing me to architect solutions that bridge abstract algorithms with real world physical constraints.

This trajectory reflects a systematic transition from treating power grids as aggregate generation-load balances to modeling them as complex computational graphs. In research recently submitted to *Sustainable Energy, Grids and Networks* (SEGAN), I established and validated a peer-to-peer trading framework that subdivides large-scale distribution networks into autonomous microgrids, integrating rigorous grid physics (AC-OPF) with distributed optimization (ADMM) across both radial and ring-connected topologies. By accounting for marginal transmission losses, this architecture reduced operational costs while strictly bounding voltage levels within the  $\pm 5\%$  threshold, preventing local trading from destabilizing the wider public grid. However, this static framework lacked the temporal foresight to handle sudden local failures.

To address this vulnerability, my graduation thesis was designed to enhance grid resilience and achieve dynamic fault tolerance. Specifically, I layered Model Predictive Control (MPC) over the trading baseline to coordinate emergency power exchanges and backup diesel generation. During simulated localized outages, this temporal coordination empowered neighboring networks to assist a failing grid, reducing forced load shedding from 15 MWh down to just 3 MWh while protecting 100% of critical loads.

Ultimately, resolving complex convergence errors has built my persistence in strict code debugging. Beyond independent problem-solving, I effectively coordinate with multiple advisors and research teams. Equipped with a robust technical toolkit (Python, Pyomo), I am ready to deploy this foundation within your ecosystem to drive scalable, real-time grid operations.

## 3. Research Interests

Building on my research experience, my focus comprises two primary research directions that shape my academic trajectory:

Currently, I focus on ensuring microgrid resilience through mathematical optimization. The theoretical power systems knowledge acquired during my academic studies serves as the foundation to accurately model physical grid constraints. Meanwhile, the optimization techniques I honed in the research laboratory, such as MILP, SOCP, and MPC, act as the practical bridge to solve complex operational challenges.

My future research expands into advanced artificial intelligence, specifically Multi-agent Reinforcement

Learning, for autonomous energy management. I aim to leverage these learning-based policies as computational accelerators, enabling real-time dispatch across decentralized grids. Crucially, this AI-driven framework provides a robust mechanism to overcome modeling inaccuracies and strict mathematical assumptions in classical solvers.

Ultimately, rather than pursuing these domains in isolation, I intend to integrate them into a mutually reinforcing architecture. My overarching objective is to leverage this synergy between physics-based modeling and data-driven learning to architect certifiably safe, autonomous power systems capable of adapting to highly dynamic and unpredictable grid conditions.

#### **4. Academic and Research Alignment**

Realizing that architecting this synergistic framework requires an ecosystem that seamlessly bridges advanced machine learning with rigorous mathematical optimization, I found the Ph.D. program at UM-Dearborn to be the perfect catalyst for my research. The interdisciplinary EECE curriculum will provide the theoretical toolkit necessary for this architecture through advanced courses such as IMSE 505 (Optimization) and ECE 5001.

Within this dynamic ecosystem, my research trajectory strongly aligns with the ongoing work of Prof. Van-Hai Bui. His pioneering research on FairMarket-RL and XRL-LLM directly addresses the integration of AI into multi-agent energy trading. By anchoring my research in his laboratory, my strong foundation in classical optimization will allow me to architect the deterministic AC safety filters required to support his learning-based policies. Prof. Bui's profound mentorship in AI surrogates is exactly what I need to bring my "synergy" concept to life.

Furthermore, my two-pillar research approach perfectly complements the broader collaborative network within the ECE department. Specifically, the groundbreaking work of Dr. Wencong Su—particularly his 2025 ECCE paper co-authored with Dr. Bui on utilizing LLMs for the Economic Dispatch problem—represents the exact application space where my ground-truth optimization models can generate indispensable training datasets. Concurrently, Dr. Junho Hong's focus on Physics-Informed anomaly detection and digital substations resonates deeply with my drive to ensure physical grid resilience. By drawing upon the LLM innovations of Dr. Su and the anomaly detection frameworks of Dr. Hong, while remaining anchored under Prof. Bui's mentorship, I will be uniquely positioned to architect a resilient, autonomous smart grid.

#### **5. Career Objectives**

Following the completion of my Ph.D, I intend to deepen my expertise as an R&D scientist within the U.S. energy sector before ultimately returning to Vietnam to contribute to the technical modernization of our national grid infrastructure. By repatriating advanced learning-based control methodologies, my long-term vision is to help transition our power architectures toward autonomous, resilient networks capable of stabilizing the rapid integration of clean energy sources. Ultimately, the Ph.D. program at the University of Michigan-Dearborn provides the exact interdisciplinary ecosystem and mentorship necessary to launch this career trajectory.